

Optimal Control of Photovoltaic Battery Storage

Ajith George¹ Ashna Benny¹ Umair Ahmad¹

¹Friedrich-Alexander Universität Erlangen-Nürnberg

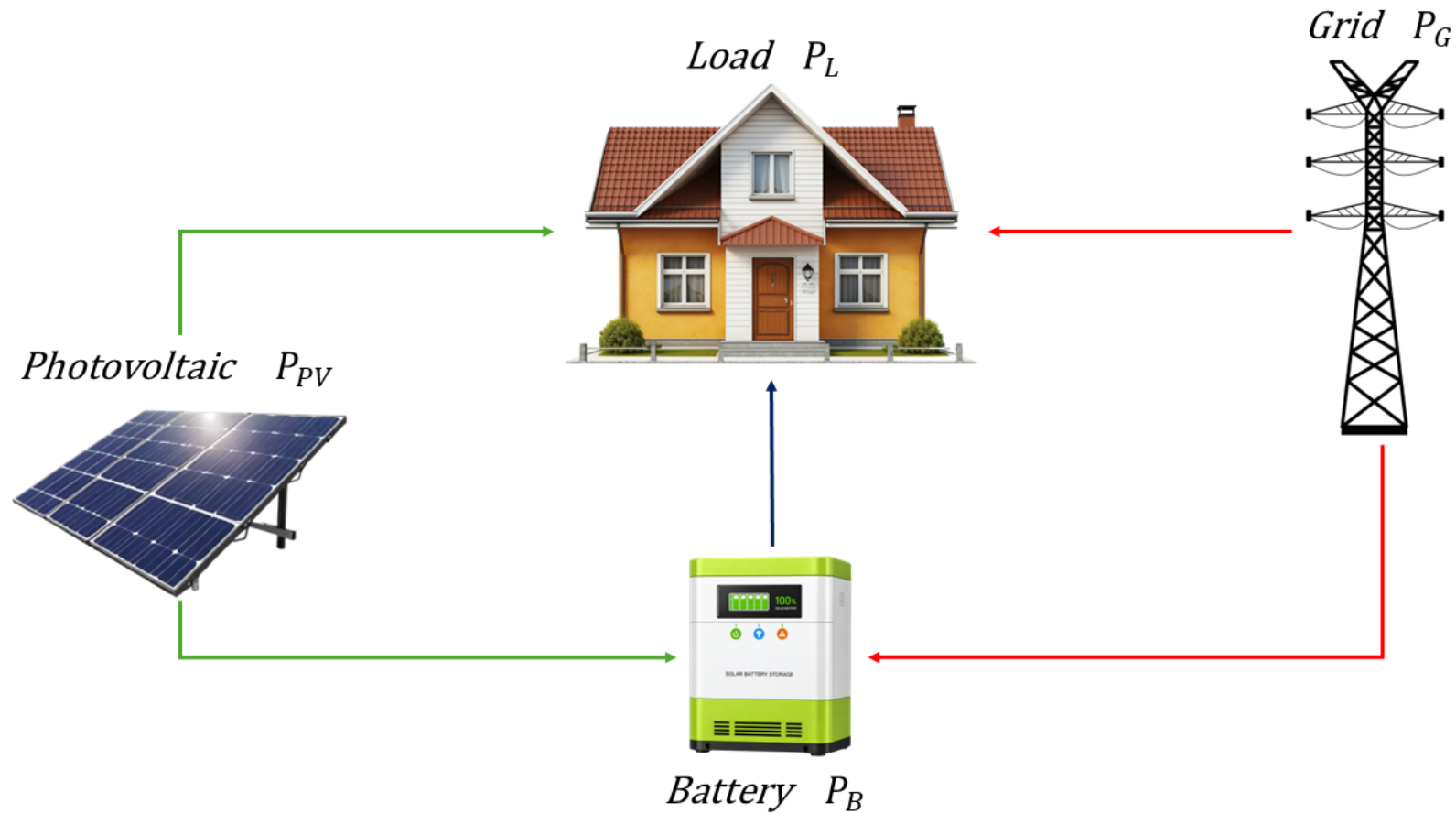
May 8, 2026

Motivation

- Solar energy is expanding worldwide
- Solar generation is intermittent
- Mismatch in demand and generation
- Need for efficient energy management
- Battery storage enables energy balancing
- Goal: Reduce electricity cost

System Description

System Components & Energy Flow Concept



Time Discretization

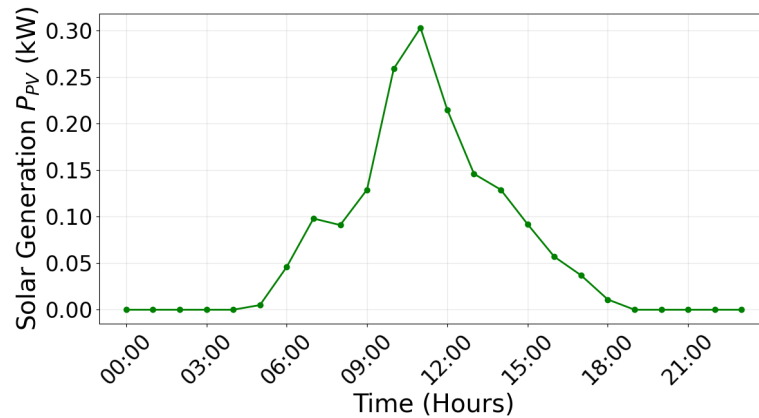
- Time horizon: *24 hours*
- Time interval : *60 minutes*
- Total steps : $i = 1, 2, \dots, 24$
- Converts continuous problem into a **finite-dimensional optimization problem**

Initial Modeling Approach

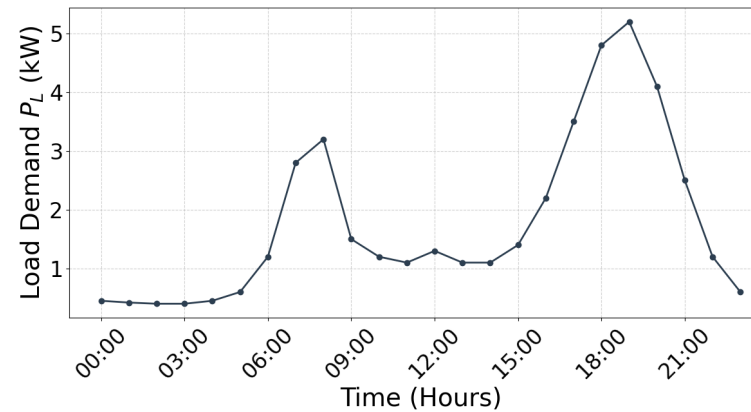
Model Inputs

- $P_{PV}(i)$: Solar generation
- $P_L(i)$: Load demand
- $P_C(i)$: Electricity price

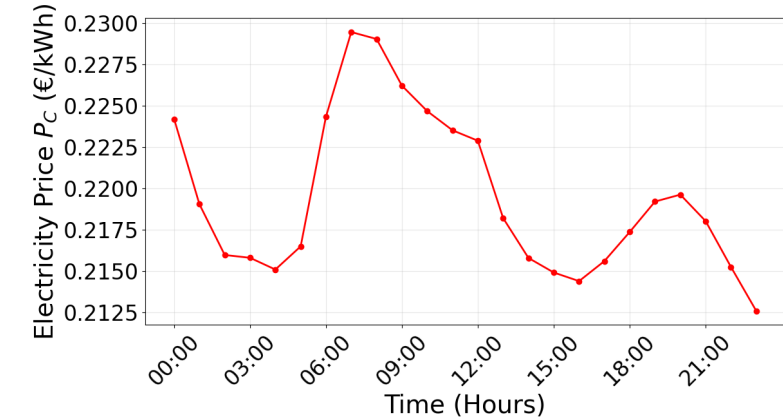
(All assumed **known from forecasts**)



Solar generation $P_{PV}(i)$



Load demand $P_L(i)$



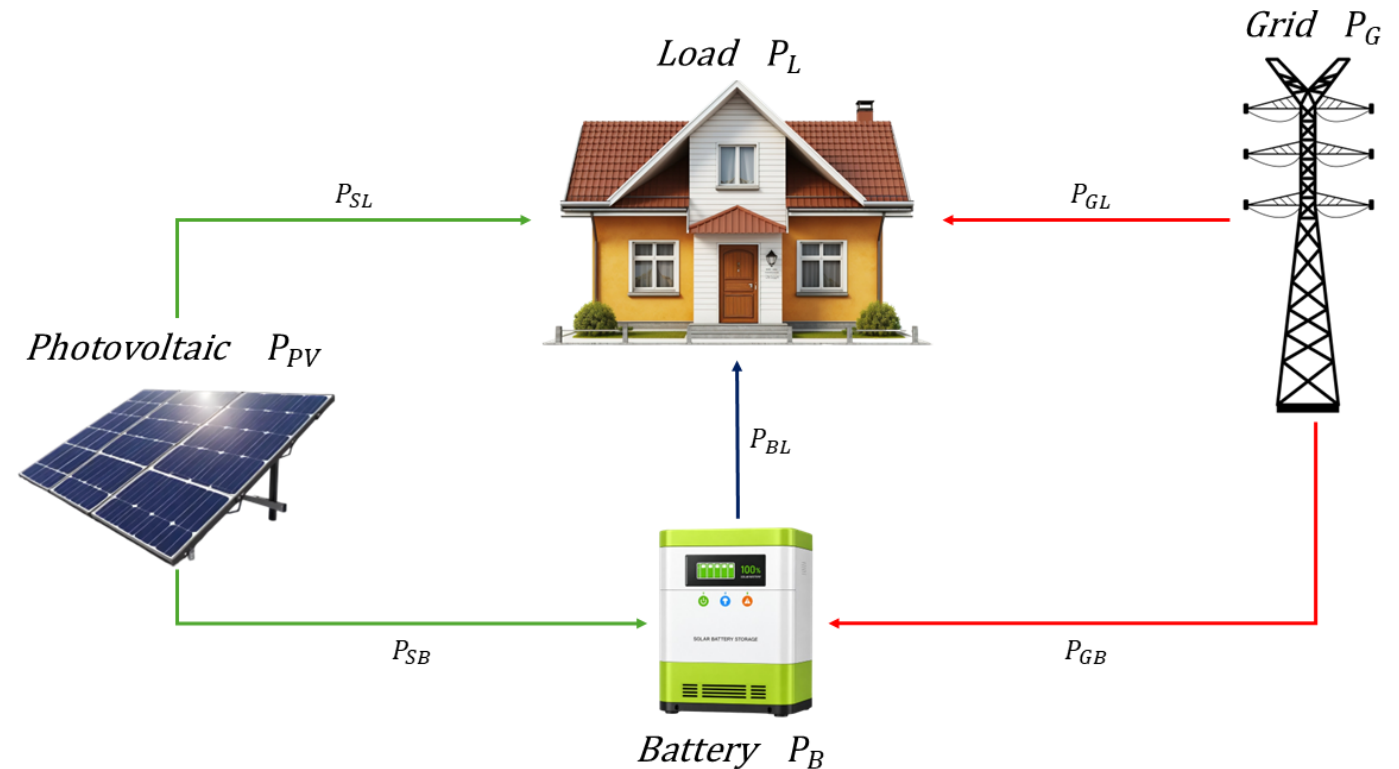
Electricity price $P_C(i)$

Data sources: PV generation from Renewables.ninja; electricity price data from ENTSO-E Transparency Platform; load profile assumed (synthetic). Data corresponds to May 8, 2019.

Initial Modeling Approach

Decision Variables

- Grid Supplies:
 - $P_{GB}(i) : \text{Grid} \rightarrow \text{Battery}$
 - $P_{GL}(i) : \text{Grid} \rightarrow \text{Load}$
- Battery supplies:
 - $P_{BL}(i) : \text{Battery} \rightarrow \text{Load}$
- Photovoltaic Supplies:
 - $P_{SL}(i) : \text{Solar} \rightarrow \text{Load}$
 - $P_{SB}(i) : \text{Solar} \rightarrow \text{Battery}$



Constraints

- Power Balance (Load Satisfaction):

$$P_{BL}(i) + P_{GL}(i) + P_{SL}(i) = P_L(i)$$

- Solar Generation Constraint:

$$P_{SB}(i) + P_{SL}(i) \leq P_{PV}(i)$$

- Battery Capacity:

$$B_{min} \leq B(i) \leq B_{max}$$

- Battery Dynamics:

$$B(i+1) = B(i) + P_{GB}(i) + P_{SB}(i) - P_{BL}(i)$$

$$\forall i = 1, 2, \dots, 24$$

Full Model

- Minimize total electricity cost
- Reduce expensive grid usage

Optimization Problem:

$$\min_{P_{BL}, P_{SL}, P_{GB}, P_{SB}, [P_{GL}]} \sum_{i=1}^{24} P_C(i) \cdot (P_{GL}(i) + P_{GB}(i))$$

such that

$$P_{BL}(i) + P_{GL}(i) + P_{SL}(i) = P_L(i)$$

$$P_{SB}(i) + P_{SL}(i) \leq P_{PV}(i)$$

$$B_{min} \leq B(i) \leq B_{max}$$

$$B(i+1) = B(i) + P_{GB}(i) + P_{SB}(i) - P_{BL}(i)$$

$$\forall i = 1, 2, \dots, 24$$

Given:

$$P_L, P_{PV}, P_C \in \mathbb{R}^{24} \text{ and } B_{min}, B_{max} \in \mathbb{R}$$

Model Type

- Deterministic optimization model
- Linear formulation
- Inputs assumed known (forecast-based)
- Finite-dimensional problem

- Phase 1: Modeling & Formulation
 - System definition and model setup
 - Deterministic linear model
- Phase 2: Solution
 - Implement in Python
 - Solve optimization problem
- Phase 3: Analysis
 - Battery behavior, grid vs solar usage
 - Cost evaluation
- Phase 4: Extension
 - Stochastic optimization
 - Robust optimization
- Phase 5: Evaluation
 - Compare results
 - Discuss improvements